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Panda iControl3

Operating Manual

Open-loop and closed-loop control system for Fischer Panda generators

Panda iControl3_eng.R01

20.10.22



Current revision status

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Revision	Page
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Hardware

Generator	Revision	Modification Strike Plate	Date	Upgrade

Created by

Fischer Panda GmbH - Head of Technical Documentation

Otto-Hahn-Str. 40

33104 Paderborn - Germany

Phone: +49 (0) 5254-9202-0

E-mail: info@fischerpanda.de

web: www.fischerpanda.de

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Fischer Panda GmbH
Otto-Hahn-Str. 40
D-33104 Paderborn
Germany

Tel. : +49 (0)5254 9202-0
Fax. : +49 (0)5254 9202-550
Hotline : +49 (0)5254 9202-767
Email : info@fischerpanda.de
Web : www.fischerpanda.de



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2. Safety instructions for the Panda iControl3

2.1 Personnel

The settings described here can be performed by the operator unless highlighted differently.

The installation should be implemented by specially trained technical personnel or by authorised workshops (Fischer Panda Service Points), only.

2.2 Safety instructions

Ensure compliance with the safety instructions in the Fischer Panda genset manual.

Note!

If these instructions are not on hand, they can be requested from Fischer Panda GmbH, 33104 Paderborn, Germany.



An external signal may trigger an automatic start-up.

Warning! Automatic start-up



The generator must not be operated with the cover removed.

Warning!



If the generator is being installed without a sound insulation capsule, it must be ensured that all rotating parts (belt pulley, belts etc.) are covered and protected so that there is no danger to life and body!

If a sound insulation capsule will be produced at the place of installation, then well-placed signs must show that the generator can only be switched on with the capsule closed.

All service, maintenance, or repair work may only be carried out when the unit is not running.

Electric voltage - DANGER TO LIFE!

Warning! Electric voltage

Electric voltages of more than 60V are potentially lethal in any situation. The rules of the respective regional authority must be adhered to for installation and maintenance.



For safety reasons, only an electrician may carry out the installation of the electrical connections of the generator.

Disconnect battery before working on the generator

Warning!

The battery must always be disconnected (first the negative terminal, then the positive terminal) if work on the generator or electrical system is to be carried out, so that the generator cannot be unintentionally started.



This applies in particular to systems with an automatic start-up function. The automatic start-up function shall be deactivated before starting work.

The flooding valve must be closed. (For PMS version only.)

Also observe the safety instructions for the other components of your system.

Note!



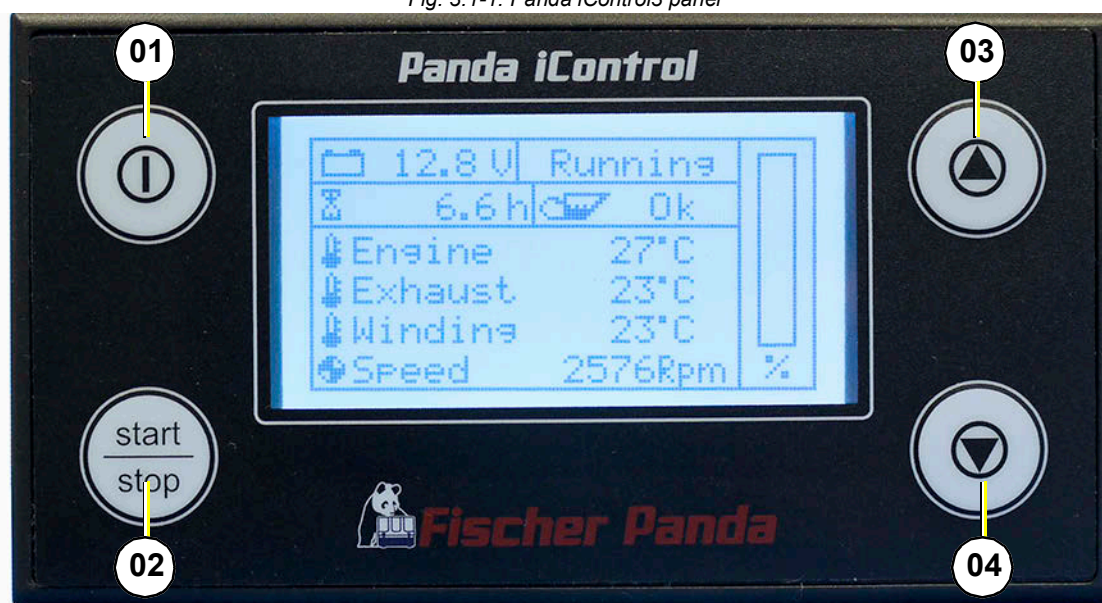
3. General operation

3.1 The Panda iControl3 panel

The "Panda iControl3 panel" control panel is the control and display unit for the Panda iControl3 control system and represents the interface between the user and the Panda iControl3 controller. The integrated display serves to present the most important data of the system as well as warnings and error messages.

The control panel is equipped with four buttons for operating the Panda iControl3 controller:

Fig. 3.1-1: Panda iControl3 panel



1. *on/off button*: Switching the Panda iControl3 controller on and off.
2. *start/stop button*: Starting and stopping the generator, confirming values in selection menus (Enter key).
3. *Cursor-up button*: Switching between display screens (up), counting values up in selection menus.
4. *Cursor-down button*: Switching between display screens (down), counting values down in selection menus.

3.2 Starting preparation / Checks (daily)

3.2.1 Marine version

1. Oil level control (ideal level: 2/3 MAX).

The level should be about 2/3 of the maximum level of a cold engine.

Further, if installed, the oil level of the oil-cooled bearing must be controlled before each start - see sediment bowl at generator front cover!.

2. State of cooling water.

The external expansion tank should be filled up to 1/3 of the maximum in a cold state. It is very important that a large expansion area remains above the cooling water level.

3. Check if sea cock for cooling water intake is open.

For safety reasons, the sea cock must be closed after the generator has been switched off. It should be re-opened before starting the generator.

4. Check raw water filter.

The raw water filter must be regularly checked and cleaned. The impeller fatigue increases, if residual affects the raw water intake.

5. Visual inspection.

Control fixing bolts, check hose connectors for leakages, control electrical connections.

6. Switch off the load.

The generator should only be started without load.

7. Open fuel valve, if installed.

8. Close battery main switch (on).

3.2.2 Vehicle version

1. Oil level control (ideal level: 2/3 MAX).

The level should be about 2/3 of the maximum level of a cold engine.

Further, if installed, the oil level of the oil-cooled bearing must be controlled before each start - see sediment bowl at generator front cover!.

2. State of cooling water.

The external expansion tank should be filled up to 1/3 of the maximum in a cold state. It is very important that a large expansion area remains above the cooling water level.

3. Visual inspection.

Control fixing bolts, check hose connectors for leakages, control electrical connections.

4. Switch off the load.

The generator should only be started without load.

5. Open fuel valve, if installed.

Close battery main switch (on).

3.3 Operation

3.3.1 Switching the controller on and off

The Panda iControl3 controller is switched on and off by briefly pressing the on/off button on the Panda iControl3 panel.

After boot-up, on the start screen information about the hardware and software release of panel and controller are displayed.

Fig. 3.3.1-1: .Panda iControl3 boot-up and start screen



Fig. 3.3.1-2: Hardware release, software release and revision of the iControl3 panel

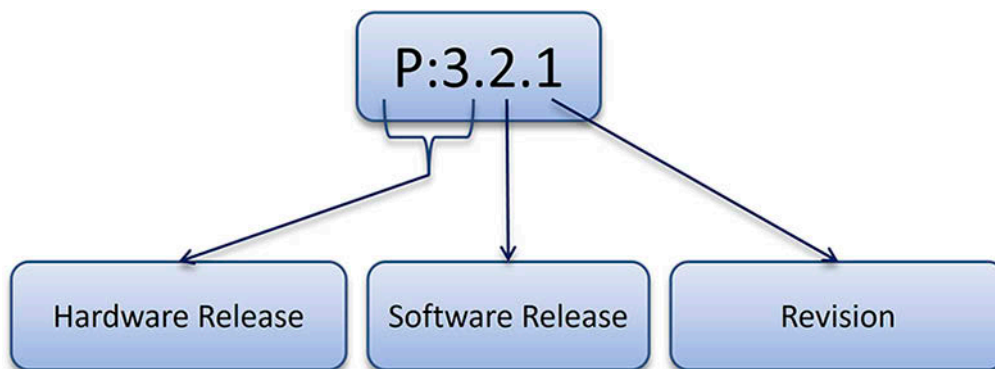
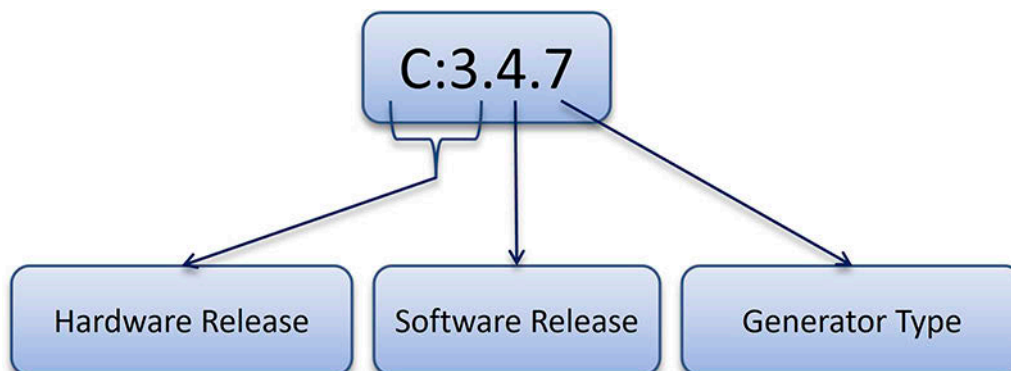


Fig. 3.3.1-3: Hardware release, software release and set generator type of the iControl3 control board



Example for the control board:
Note!

Hardware version: C:3 -> iControl3 control board

Software version: 4 -> Release 4 of the iControl3 software, compatible with iControl3 panel

Generator type: 7 -> Panda 15000i 230 V / 50 Hz PMS



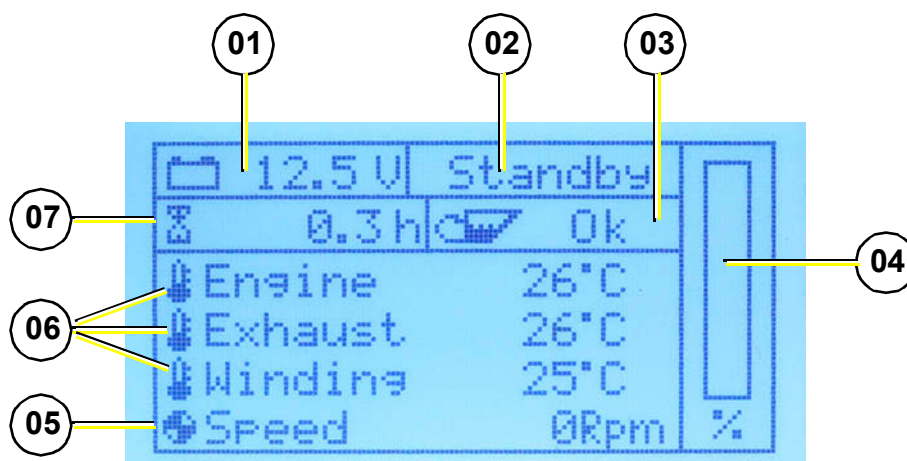
3.3.2 Default display screen

Five seconds after the controller is switched on, the display will change to the default display screen. On the default display screen, you will find information on the battery voltage, operating hours of the generator, temperatures of cylinder head, exhaust manifold, and winding, RPM, and the oil pressure status. Also, a bar graph display at the right hand edge of the display shows the utilisation of the generator in percent.

Data output on the default display screen:

- Battery voltage (supply voltage)
- Status field for operating modes (stand-by, pre-heat, starting, override, running, autostart, stopping)
- Operating hours of the generator
- Oil pressure status
- Cylinder head temperature
- Temperature of exhaust manifold
- Winding temperature
- Speed/RPM
- Utilisation in percent

Fig. 3.3.2-1: Default display screen



01. Starter Battery voltage [V]

02. Generator status

03. Oil pressure status [Ok / fault]

04. Utilisation [%]

05. Generator speed [rpm]

06. Temperatures [°C] (Engine, Exhaust and Winding)

07. Operating hours [h]

The Display shows the iControl3 board input voltage.

Note!

At generator systems with 12 V starter system this voltage is equal to the starter battery voltage.

At generator systems with 24 V starter system the starter battery voltage can not be displayed.



3.3.3 Operating modes

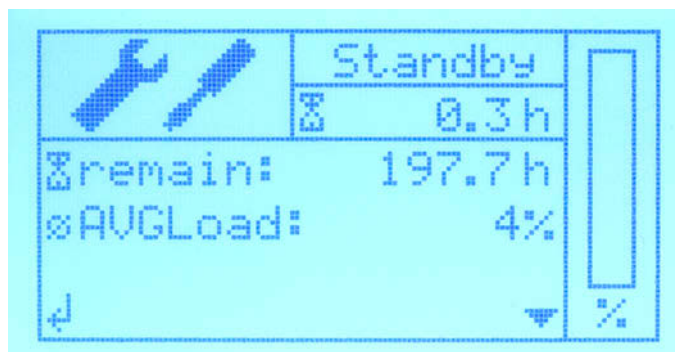
The Panda iControl3 controller offers different operating modes.

3.3.3.1 Stand-by mode

After the controller is switched on with the on/off button, the system is in stand-by mode. This is indicated by the output "Standby" in the status field in the top right corner of the default display screen.

In this operating mode, the system can be switched off with the on/off button, and the generator can be started up with the start/stop button. With the cursor buttons, the service information screen can be accessed.

Fig. 3.3.3.1-1: Service information screen



The total operating hours of the generator are given on the default display screen and on the service information screen. By actuating the cursor-up button in stand-by mode, the service screen can be accessed.

This screen is marked with a screwdriver/spanner symbol. Here, the time until the next service and the average load in percent are given. By actuating the cursor-up or cursor-down button, you can return to the default screen.

With the dynamic operation hours the service interval can be raised up to 30 % (200 h max.). Make sure that the dynamic operation hours are not reset accidentally between the service interval, **see "Resetting the service interval (»Service«)" on page 20.**

Note!



In the set-up menu of the controller, you can reset the service interval after performing maintenance, **see "Set-up menu" on page 14.**

3.3.3.2 Start-up mode

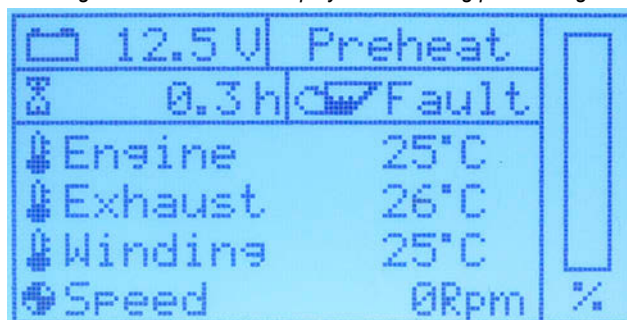
The start-up mode represents the transition from stand-by mode to operation mode, i.e., generator operation. By actuating the start/stop button in stand-by mode, you can initiate the start-up process of the generator.

The pre-heating is the first step. During this stage, the status field at the top right of the default display screen shows the word "Preheat".

The pre-heating is always implemented for a duration of 10 seconds, regardless of the cylinder head temperature.

In temperatures below 0°C, the pre-heating time is always 40 seconds.

Fig. 3.3.3.2-1: Default display screen during pre-heating



After pre-heating, the starter is initiated, accompanied by the text output "Starting" in the status field of the default display screen.

Fig. 3.3.3.2-2: Default display screen during start-up



The controller will only perform one starting attempt. If the generator could not be started, the text output "Starting fails" informs you of the failure of the generator starting attempt.

Note!



Acknowledging the message with the start/stop button on the Panda iControl3 panel will return the system to stand-by mode.

Fig. 3.3.3.2-3: Error message »Starting fails«



If there is difficulty in starting - close the seacock (Panda Marine Generators only)

Attention!



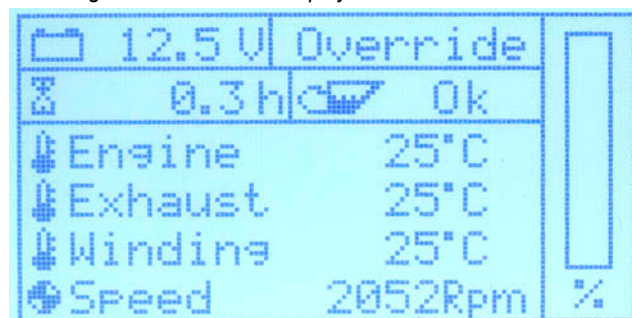
If the generator engine does not start immediately and further start attempts are necessary, then the seacock **MUST** be closed (i.e. for ventilating the fuel lines ect.) The cooling water impeller pump turns automatically and draws cooling water as long as the motor is turning. If the diesel motor is running, the cooling water is blown out by the exhaust system gases. The cooling water cannot be pressed through the exhaust as long as the diesel motor does not run at sufficient speed. This leads to severe motor damage.

Open the sea valve as soon as the generator is started.

3.3.3.3 Override mode

The override mode follows directly after the successful start-up of the generator. In this mode, no fault analysis is performed. The duration of the override mode is 10 seconds. The status indicator on the display reads "Override".

Fig. 3.3.3.3-1: Default display screen in override mode



3.3.3.4 Operation mode

Operation mode signifies the operating mode in which the generator is running and all operating data are within their normal range. The status field of the default display screen shows "Running".

In operation mode, the electrical load is given on the right hand side of the default display screen and in the inverter screen as a bar graph. The bar graph merely provides a guide value for the load of the generator and gives the values as a percentage.

Fig. 3.3.3.4-1: Default display screen in operation mode

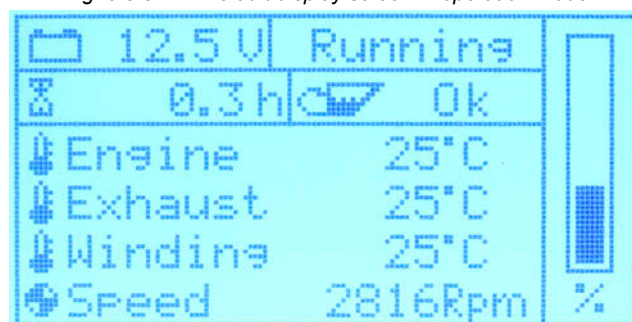
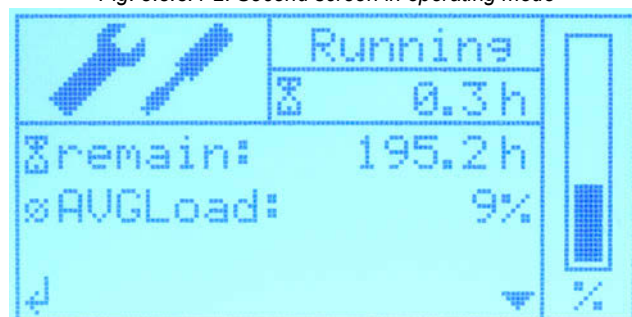


Fig. 3.3.3.4-2: Second screen in operating mode

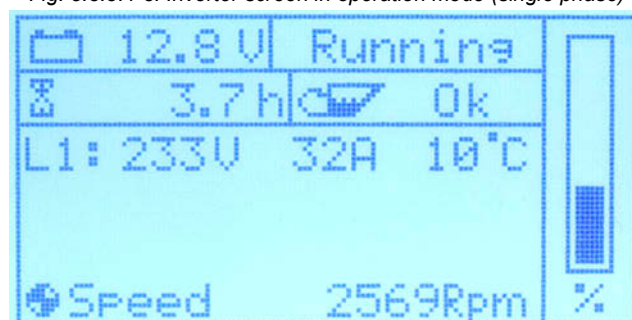


Display screen for single phase generators

With the single phase i-series generators, there is an additional screen in operation mode for the inverter data. This screen provides the current inverter output voltage, current and the inverter temperature.

You can access the inverter screen by actuating the cursor-down button while in operation mode.

Fig. 3.3.3.4-3: Inverter screen in operation mode (single phase)

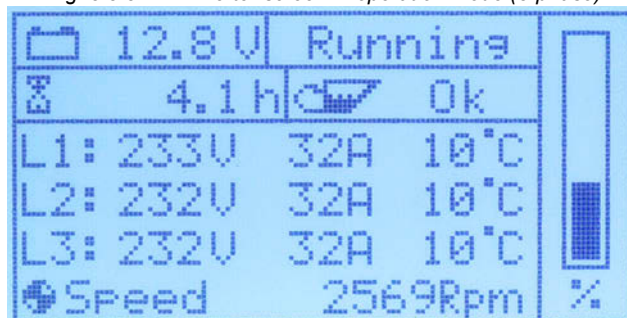


Display screen for 3-phase generators

With the 3-phase i-series generators, there is an additional screen in operation mode for the inverter data. This screen provides the latest inverter output voltage of the single phases with the matching conductor current and the circuit board temperature. The inverter will be switched off at a circuit board temperature of 75 °C.

You can access the inverter screen by actuating the cursor-down button while in operation mode.

Fig. 3.3.3.4-4: Inverter screen in operation mode (3-phase)



3.3.3.5 Panda i-Generator with electro-magnet Clutch (optional)

During the activation of the electro-magnetic clutch, the iControl3 raise the generator speed to maximum.

After the clutch is released, the generator speed will drop to normal.

Attention!



3.3.3.6 Stop mode

By activating the start/stop button in operation mode, i.e., while the generator is running, you will stop the generator. After stopping the generator, the system will return to stand-by mode. The display status field reads „Stopping“.

With marine generators and a cylinder head temperature greater than 70 °C (for example after a long time running with high load) the iControl3 starts a stopping delay timer. The Display shows „Cooldown“ in the status field.

This does not exist with vehicle generators, since the fan cannot work during cooldown.

During this time the iControl3 system will shut of the PMGi and run the engine at idle speed. During the delay time an automatic start request will be ignored.

After the delay time, the generator will be stopped automatically.

You can interrupt the delay time by pressing the start/stop button. (Not recommend by Fischer Panda. The Engine may overheat)

Never use an emergency stop switch for a regular stop of the generator.

Attention!



The engine may overheat and can be damaged/destroyed

If the generator is manually started up and stopped while in automatic start-up mode, it will switch to stand-by mode for safety reasons.

Note! Manual start in autostart mode



If necessary, the autostart mode must be reactivated.

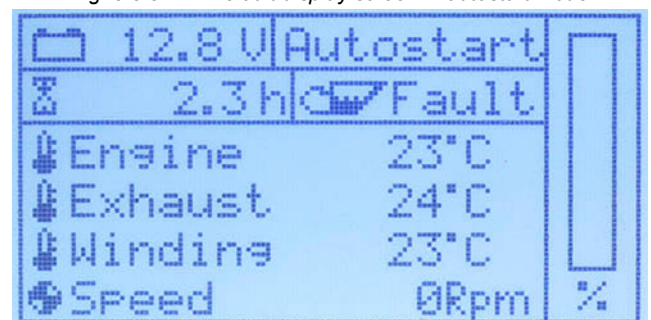
3.3.3.7 Autostart mode

The Panda iControl3 panel is equipped with an autostart function. A jumper between pin 6 (AS) and pin 7 (AS) of the Phoenix jack of the control panel starts up the generator with a delay of 5 seconds when the autostart function is active. Removing the jumper will stop the generator - also with a delay of 5 seconds.

The autostart function must be activated in the set-up menu. To activate the autostart function, see **“Activating/deactivating the autostart function (»Autostart«)”** on page 21.

The display status field reading **“Autostart”** indicates that the autostart function is active, or, if it reads **“Standby”**, this means that the autostart function was deactivated.

Fig. 3.3.3.7-1: Default display screen in autostart mode



The autostart function will remain active even after the controller is switched off and on again with the on/off button. If the autostart function is no longer required, the function must be deactivated, see **“Activating/deactivating the autostart function (»Autostart«)”** on page 21.

Warning! Automatic start-up



If the generator is manually started up and stopped while in automatic start-up mode, it will switch to stand-by mode for safety reasons.

Note! Manual start in autostart mode



If necessary, the autostart mode must be reactivated.

3.4 Set-up menu

In the set-up menu, a series of parameters concerning panel, generator and service topics can be modified directly using the control panel.

The set-up menu consists of 3 submenus and an option to save changes.

Fig. 3.4-1: Submenus of the set-up menu

Menu item	Settings range for
Panel	Parameters concerning the iControl3 panel
Generator	Parameters concerning the generator control iControl3
Service	Parameters concerning service topics
Save	Saving the values and restarting the iControl3

Calling up the set-up menu:

1. Switch on the iControl3 system using the on/off button.
2. Press the cursor-up button **while the hardware/software versions are being displayed on the start screen.**

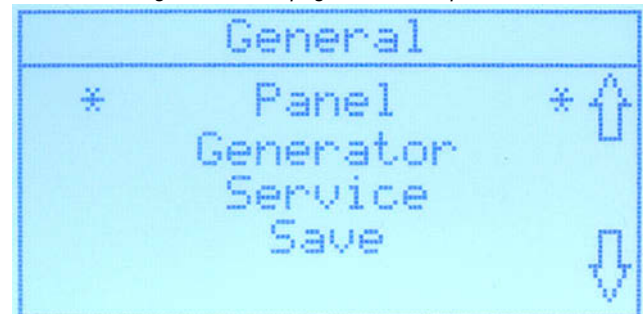
Fig. 3.4-2: Versions are being displayed on the start screen



The start page of the set-up menu is displayed with its submenus:

- Panel
- Generator
- Service
- Save

Fig. 3.4-3: Start page of the set-up menu



The submenu »**Service**« is blocked and can only be accessed by trained personnel and Fischer Panda employees.

Note!



In all submenus, set the desired values using the cursor buttons, then confirm your settings with the start/stop button.

Note!



After setting all parameters, you can exit the set-up menu with the menu item "Save". In doing so, all settings entered in the sub-menus Backlight 1, Backlight 2, Dimtime, Language, Units and Network ID are saved.

After saving the changes, the iControl3 restarts and the start screen is displayed. After that all changes will take effect.

3.4.1 Submenu »Panel«

Calling up the submenu »Panel«:

- Select *Panel* and confirm with the start/stop button.

Fig. 3.4.1-1: Set-up menu, highlighted *Panel*



With the cursor-up and cursor-down buttons, you can navigate through the menu. The currently selected menu item is marked with two asterisks (*), e. g. *Backlight 1*.

Fig. 3.4.1-2: Submenu »Panel« highlighted *Backlight 1*



Set-up submenu »Panel« with item highlighted: *Back*

Fig. 3.4.1-3: Submenu »Panel«, highlighted *Back*



The start/stop button is used to confirm the selected item in the set-up menu. Confirming the row marked with the * accesses the selected submenu.

Note!



Fig. 3.4.1-4: Items of the submenu »Panel«

Menu item	Settings range for
Backlight 1	Setting the brightness value for the standard backlighting to 0-100 %
Backlight 2	Setting the brightness value for the dimmed backlighting to 0-100 %
Dimtime (dimming time)	Time until the display switches to dimmed mode, 0-255 s (corresponds to 0-4 min), 0= lighting is always dimmed
Language	Language selection (English, German and French available)
Units	Switches the display between °C and °F
Network	Settings for the network ID of the panel
Back	Exiting the set-up submenu »Panel«

3.4.1.1 Setting the backlight (»Backlight« and »Dimtime«)

The brightness of the display backlight of the Panda iControl3 panel can be varied from 0 to 100 % in 1 % steps. Also, the display can be dimmed with a timer if no button is actuated on the control panel for a parameter is able

period. To adjust the default brightness and the dimmed brightness, the set-up submenu »Panel« offers the items "Backlight 1" (default brightness) and "Backlight 2" (dimmed brightness). These service menu screens are highlighted with the light bulb symbol:



- Select *Backlight 1* and confirm with the start/stop button.
- Enter the value for the backlight. Values between 0 and 100 % are possible.

Fig. 3.4.1.1-1: Submenu »Panel«, highlighted *Backlight 1*



Fig. 3.4.1-2: Screen for setting the brightness of the display backlight



01. Increase the value (Cursor-up button) [%]

02. Decrease the value (Cursor-down button) [%]

03. Set value [%]

04. Confirm the set value (start/stop button)

Setting the value for the dimmed brightness (Backlight 2) works in the same way.

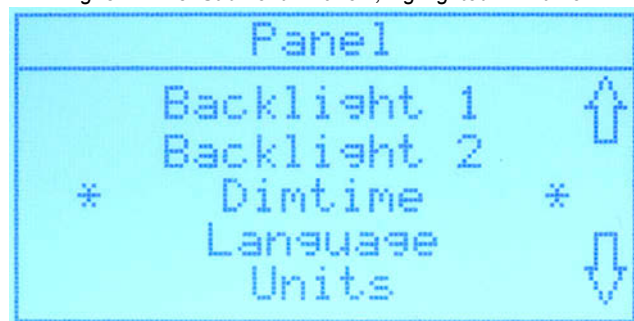
Note!



The period after which the backlight switches to the dimmed level can be specified with the menu item "Dimtime".

- Select *Dimtime* and confirm with the start/stop button.

Fig. 3.4.1.1-3: Submenu »Panel«, highlighted *Dimtime*



- Enter the value for the dimtime. Values between 0 and 255 s are possible.

Fig. 3.4.1.1-4: Screen for setting the dimtime



3.4.1.2 Setting the language («Language«)

- Select *Language* and confirm with the start/stop button.

Fig. 3.4.1.2-1: Submenu »Panel«, highlighted *Language*

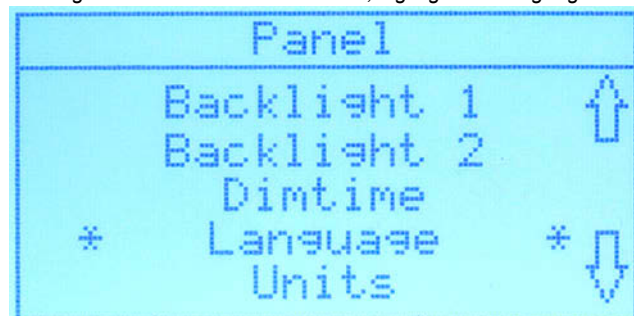
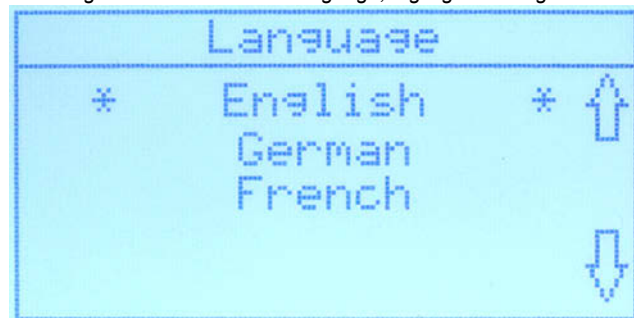


Fig. 3.4.1.2-2: Selected language, highlighted *English*



Settings options:

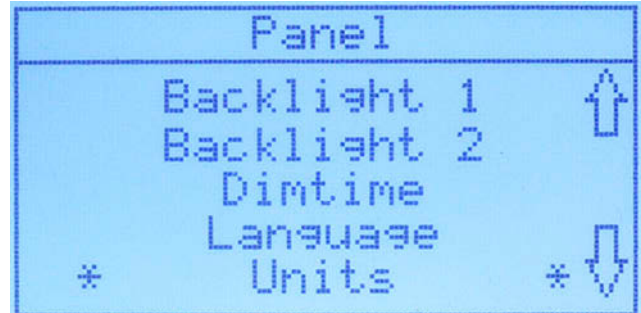
- **English**
- German (Not yet available. Planned in the future)
- French (Not yet available. Planned in the future)

3.4.1.3 Selecting the unit for temperature value outputs (»Units«)

Temperature values can be displayed in degrees Celsius [°C] or in degrees Fahrenheit [°F].

- Select ***Units*** and confirm with the start/stop button.

Fig. 3.4.1.3-1: Submenu »Panel«, highlighted ***Units***

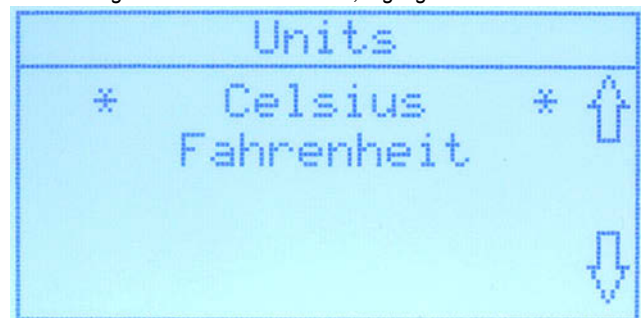


Settings options:

Celsius: Output of all temperatures in degrees Celsius [°C]

Fahrenheit: Output of all temperatures in degrees Fahrenheit [°F]

Fig. 3.4.1.3-2: Selected unit, highlighted ***Celsius***



3.4.1.4 Setting the network ID

Settings in this area must only be entered by Fischer Panda associates and Fischer Panda service points.

Changing the network ID can result in malfunction.

- Select ***Network*** and confirm with the start/stop button.

Stop! Network ID must not be modified

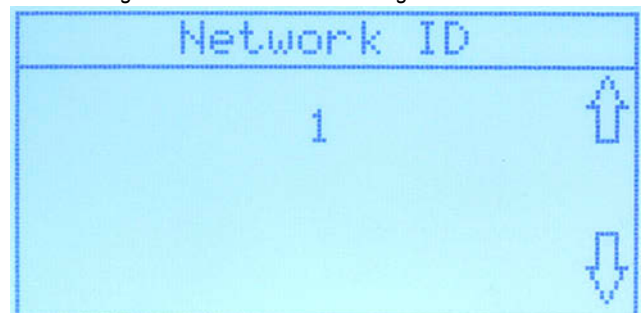


Fig. 3.4.1.4-1: Submenu »Panel«, highlighted ***Network***



- Enter the network ID.

Fig. 3.4.1.4-2: Screen for setting the network ID



3.4.2 Submenu »Generator«

Call up the submenu »Generator«:

- Select *Generator* and confirm with the start/stop button.

Fig. 3.4.2-1: Set-up menu, highlighted *Generator*



With the cursor-up and cursor-down buttons, you can navigate through the menu. The currently selected menu item is marked with two asterisks (*), e. g. *Prime fuel*.

Fig. 3.4.2-2: Submenu »Generator«, highlighted *Prime fuel*

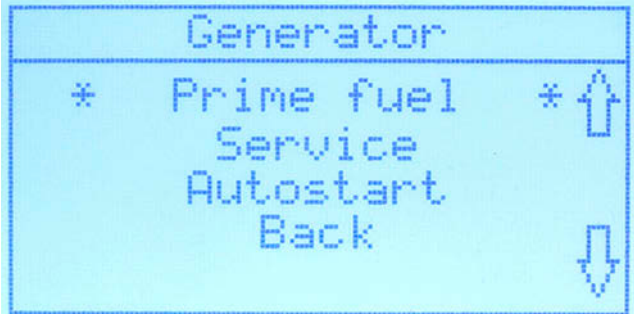
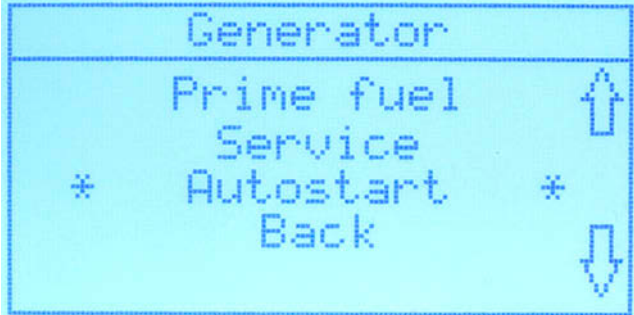


Fig. 3.4.2-3: Submenu »Generator«, highlighted *Autostart*



Set-up submenu »Generator« with item highlighted: *Autostart*

The start/stop button is used to confirm the selected item in the set-up menu. Confirming the row marked with the * accesses the selected submenu.

Note!



Fig. 3.4.2-4: Items of the submenu »Generator«

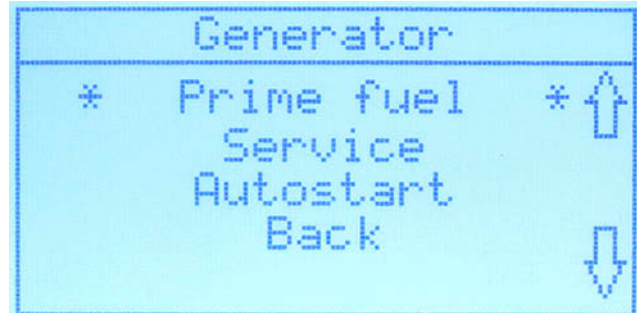
Menu item	Settings range for
Prime fuel	Activation of the fuel pump to prime the generator fuel system
Service	Resetting the "Operating hours to service" indication
Autostart	Activating and deactivating the automatic start-up function
Config	Password protected area for Fischer Panda associates and Fischer Panda service points
Back	Exiting the set-up submenu »Generator«

3.4.2.1 Priming the fuel system («Prime fuel«)

To prime the fuel system, the Panda iControl3 offers the option of separately activating the fuel pump. In the set-up submenu »Generator«, select the menu item "Prime fuel" and confirm your selection using the start/stop button.

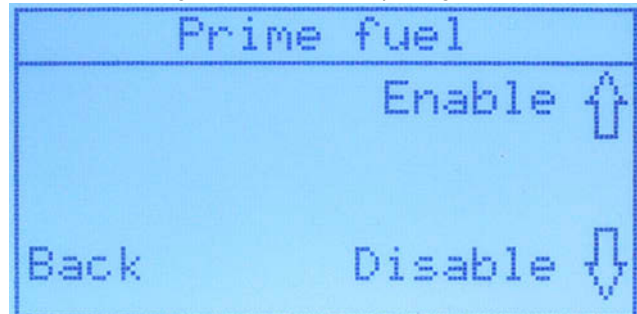
- Select *Prime fuel* and confirm with the start/stop button.

Fig. 3.4.2-1: Submenu »Generator«, highlighted *Prime fuel*



- Actuate the cursor-up button (Enable) to switch the fuel pump on.
- Actuate the cursor-down button (Disable) to switch the fuel pump off.
- Actuate the start/stop button (Back) to go back to set-up submenu »Generator«.

Fig. 3.4.2-2: Screen for priming fuel



3.4.2.2 Resetting the service interval («Service«)

As the indication of operating hours remaining until the next service interval can be reset at any time, it serves only as an orientation guide. The service intervals shall be implemented using the actual operating hours and shall be properly documented in the service log of the generator.

Note!



With the dynamic operation hours the service interval can be raised up to 30 % (200 h max.). Make sure that the dynamic operation hours are not reset accidentally between the service interval.

Note!



- Select *Service* and confirm with the start/stop button.

Fig. 3.4.2.2-1: Submenu »Generator«, highlighted *Service*

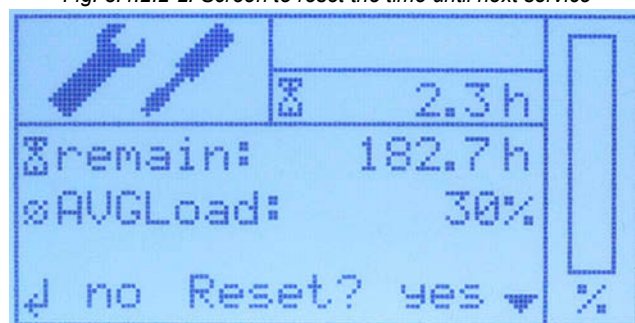


- Actuate the cursor-down button (yes) to reset the service interval.
- Actuate the start/stop button (no) to aborts the process.

The service interval for each generator type is stored in the software.

The iControl3 restarts after resetting the service interval. After restart, the new value will be displayed in the service screen.

Fig. 3.4.2.2-2: Screen to reset the time until next service



3.4.2.3 Activating/deactivating the autostart function («Autostart»)

DANGER TO LIFE! - Improper operation can result in health impairment and death.

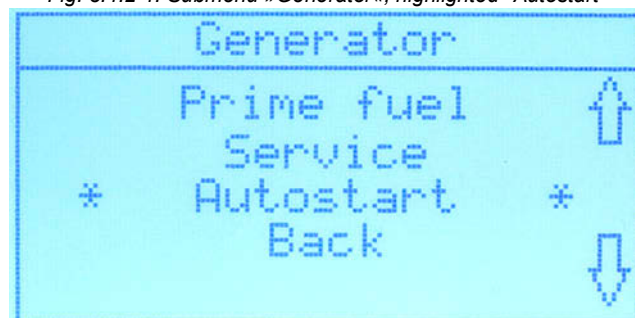
Warning! Automatic start-up



While the automatic start-up function is active, the generator can start up automatically. Before activating it, ensure that the generator capsule is closed and that the corresponding warning signs are affixed to the generator.

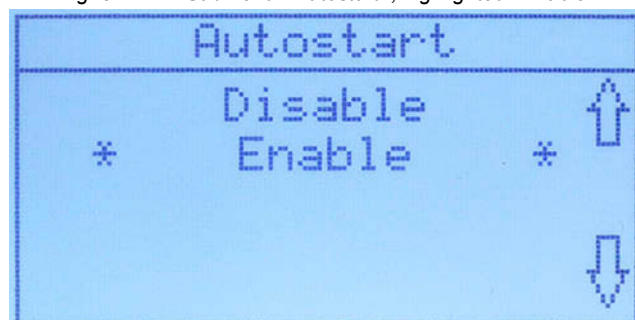
- Select *Autostart* and confirm with the start/stop button.

Fig. 3.4.2-1: Submenu »Generator«, highlighted *Autostart*



In the "Autostart" submenu, you can select between the options "Disable" and "Enable" using the cursor buttons:

Fig. 3.4.2-2: Submenu »Autostart«, highlighted *Enable*



To activate the autostart function, select "Enable" and confirm with the start/stop button.

To deactivate the function, select the menu item "Disable" and confirm with the start/stop button.

After confirming "Enable"/"Disable" iControl3 goes back to the set-up submenu »Generator«.

Fig. 3.4.2-3: Set-up submenu »Generator« highlighted *Back*



Complete activation/deactivation:

- Select *Back* and confirm.

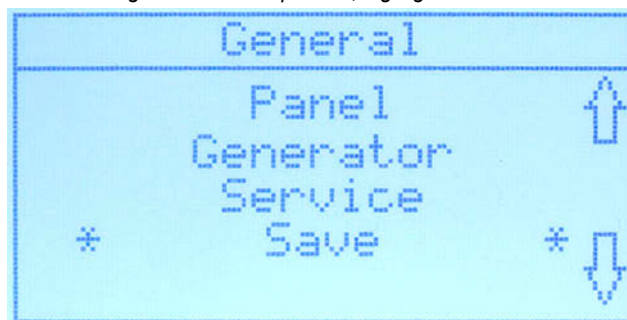
iControl3 goes back to the general set-up menu.

Save changes:

- Select *Save* and confirm.

Result: The iControl3 restarts.

Fig. 3.4.2-4: Set-up menu, highlighted *Save*



After the restart, the autostart function is activated/deactivated.

The display status field reading "Autostart" indicates that the autostart function is active.

If it reads "Standby", the autostart function was deactivated.

Fig. 3.4.2-5: Default display screen in autostart mode



The autostart function will remain active even after the controller is switched off and on again with the on/off button. If the autostart function is no longer required, the function must be deactivated with "Disable" as described above.

The autostart function of the Panda iControl3 is now ready. While the autostart function is active, you can manually start and stop the generator with the start/stop button, as well.

If the generator is manually started up and stopped while in automatic start mode, it will switch to stand-by mode for safety reasons.

Warning! Automatic start-up



Note! Manual start in autostart mode



If necessary, the autostart mode must be reactivated.

3.4.3 Submenu »Save«

After setting all parameters, you can exit the set-up menu with the menu item "Save".

- Select *Save* and confirm with the start/stop button.

All settings entered in the sub-menus Panel, Generator and Service are saved.

Fig. 3.4.3-1: Text



After saving the changes, the iControl3 restarts and the start screen is displayed. After that all changes will take effect.

Note!:



3.5 iControl3 Emergency Stop

The iControl3 is prepared for the connection of an emergency stop. The socket for the emergency stop is in the cable harness integrated and bridged (1X1, optional emergency off). The bridge must be removed and the emergency stop connected.

Fig. 3.5-1: Emergency stop bridge in the cable harness



After the emergency stop is initiated, the servo drives to zero position, all outputs of the iControl3 controller are switched off and the power supply to the inverter is switched off.

The Panel shows „Emergency Stop“. The message disappears as soon as the emergency switch is released.

Fig. 3.5-2: Error message »Emergency stop«



4. Installation

All connecting wires and instructions for installation are designed and adequate for „standard“ installation situations.

Warning! Properly dimension your system.



As Fischer Panda does not know the specific installation and operating situation (e.g. special vehicle shapes, high travel speeds, and special application conditions, etc.), this installation specification can only serve as a guideline and example. The installation must be adjusted and implemented by a competent specialist based on the local conditions and requirements.

If damage occurs due to wrong installation without adjusting for specific conditions, it is not covered by the warranty.

4.1 Personnel

The installation described herein must be implemented by specially trained technical personnel or by authorised workshops (Fischer Panda Service Points), only.

4.1.1 Hazard warnings for installation

Ensure compliance with the general safety instructions at the beginning of this manual.

Note!



DANGER TO LIFE! - Improper operation can result in health impairment and death.

Warning! Automatic start-up



The battery must always be disconnected (first the negative terminal, then the positive terminal) if work on the generator or electrical system is to be carried out, so that the generator cannot be unintentionally started.

Improper installation can cause severe injury and/or substantial property damage. Therefore:

Warning! Risk of injury!



- Always turn off motor to perform installation work.
- Ensure adequate space for assembly prior to starting work.
- Ensure order and cleanliness at the work place! Parts and tools loosely stacked or lying on the floor represent accident hazards.
- Use only standard tools and special tools for installation work. Incorrect or damaged tooling can result in injury.

DANGER TO LIFE! - Improper operation can result in health impairment and death.

Warning! Electric voltage



Electric voltages of more than 60 V are potentially lethal in any situation. The rules of the respective regional authority must be adhered to during installation. For safety reasons,

only an electrician may carry out the installation of the electrical connections of the generator.

Generator and cooling water may be hot during and after operation. Burn/scalding hazard!

During operation, overpressure may build up in the cooling system.

For installation work, personal protective equipment is compulsory. This includes:

- Tightly fitting protective clothing
- Safety shoes
- Safety gloves
- Hearing protection
- Safety goggles if applicable

All loads must be disconnected prior to working on the generator to avoid damage to the devices.

Warning! Hot surface/material



Mandatory instruction! Protective equipment required



Warning! Switch off all loads.



4.2 Disposal of the components

Electronics components are hazardous to the environment and contain rare raw materials.

Collect and properly dispose of components that are no longer needed!

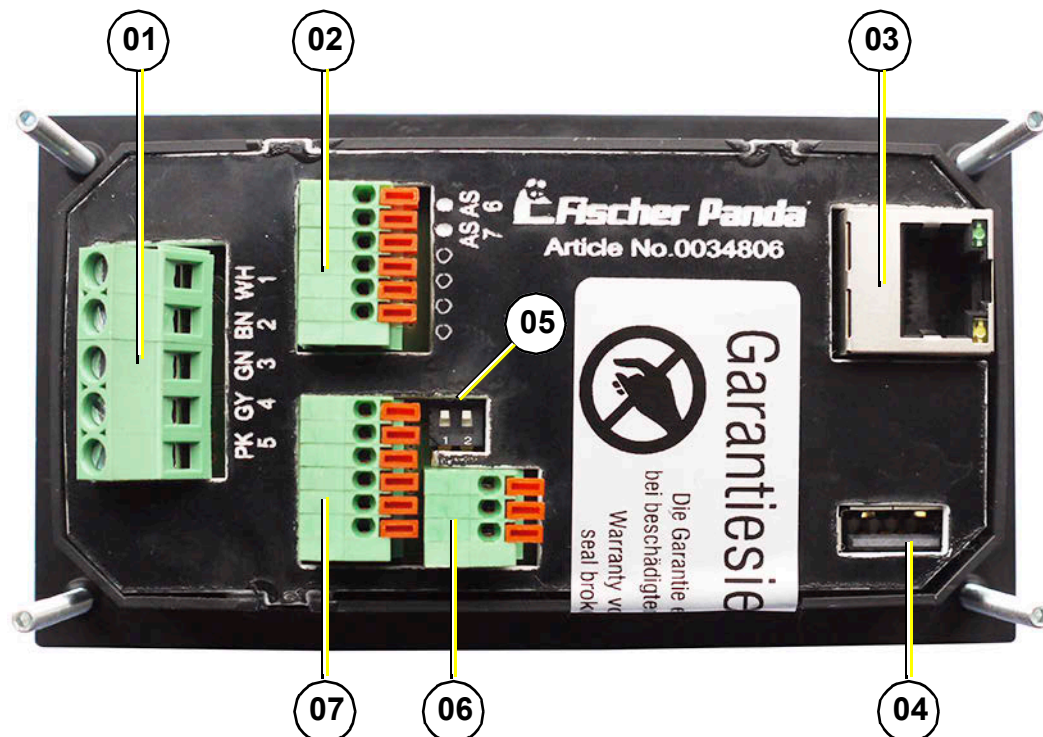
Mandatory instruction! Protect the environment.



The iControl3 board is typically pre-installed on the generator, and the corresponding connecting lines for connecting it to the iControl3 panel and the PMGi are prepared. See generator manual.

4.2.1 Panda iControl3 panel with installation housing

Fig. 4.2.1-1: Panda iControl3 panel, backside with connectors, DIP switch and closed housing



4.2.2 Connectors and switches on the Panda iControl3 panel

There are 6 connectors and a double DIP switch on the backside of the iControl3 panel.

Fig. 4.2.2-1: Connectors and DIP switch of the iControl3 panel

Connector	Function
1	5-pin Phoenix socket for panel connecting cable
2	Automatic start (pins AS 6 and AS 7)
3	Network, for example Ethernet (for future use)
4	USB (for future use)
5	Double DIP switch, switching on/off the terminating resistors for CAN bus 1 and CAN bus 2
6	Fischer Panda CAN Bus 2, CAN interface, j1939 based »fpCAN« (for future use)
7	3 universal outputs (for future use)

4.2.2.1 Terminal assignment of the 5-pin Phoenix socket

The Panda iControl3 panel is connected via a 5-pin Phoenix socket.

Fig. 4.2.2.1-1: Terminal assignment on the Panda iControl3 panel

Terminal	Terminal description	Cable colour	Function
1	UBUS	white (WH)	Bus supply voltage
2	GND	brown (BN) + shielding	Fischer Panda bus ground, ground connection between Panda iControl3 control unit and Panda iControl3 panel
3	RZL	green (GN)	Exciter wire, is switched to ground if the control unit is to switch on
4	DATA-A	grey (GY)	Fischer Panda CAN bus 1
5	DATA-B	pink (PK)	Fischer Panda CAN bus 1

Use only original Fischer Panda connecting cables.

Note!



4.2.2.2 Terminal assignment of the automatic start socket

The automatic start contact of the Panda iControl3 panel is located on a separate 6-pin socket. A jumper between terminal 6 and 7 closes the autostart contact.

Fig. 4.2.2.2-1: Terminal assignment of the automatic start socket

Terminal	Terminal description	Cable colour	Function
6	UBATT	--	Automatic start
7	USTART/STOP	--	Automatic start

4.2.2.3 Terminal assignment Fischer Panda CAN-Bus 2

Second CAN interface, j1939 based »fpCAN«.

Note! For future use.



Fig. 4.2.2.3-1: Terminal assignment of the second CAN interface

Terminal	Terminal description	Cable colour	Function
1	CAN		
2	CAN		
3			

4.2.2.4 Function of the double DIP Switch

The terminating resistors for CAN bus 1 and CAN bus 2 are switched on and off with the double DIP switch.

Notice the information on the operation of master and slave panels. See “Master and Slave Panels” on page 34.

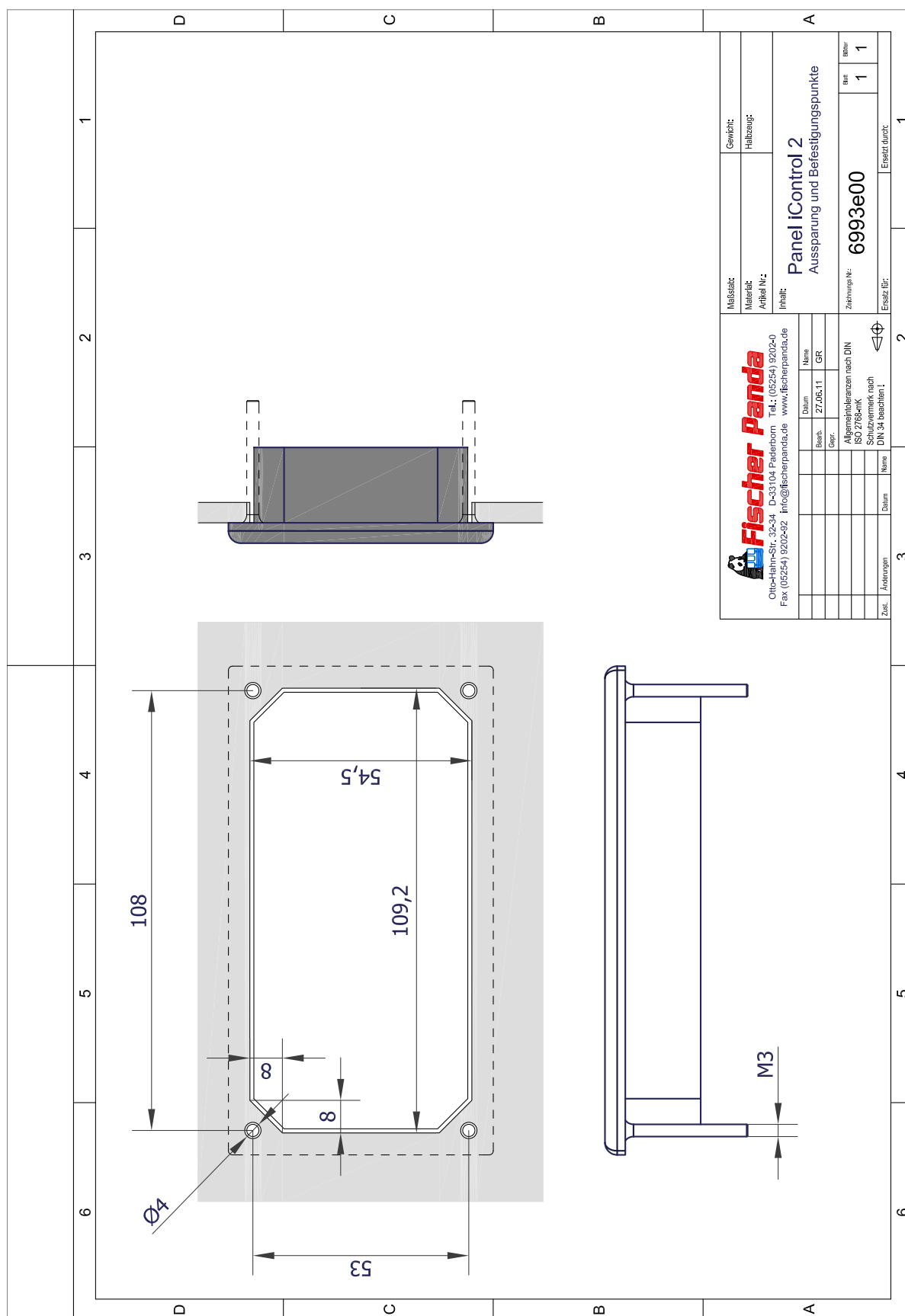
Note!



Fig. 4.2.2.4-1: Positions of the double DIP switch

Switch	Function	Switch position
1	Terminating resistor for CAN bus 1 On/Off	top - terminating resistor “ ON ”, down - terminating resistor “ OFF ”
2	Terminating resistor for CAN bus 2 On/Off	

Fig. 4.3-1: Housing of the Panda iControl3 panel



Due to the terminals being exposed, the protection class IP 04 applies to the iControl3 panel. Please note!

If properly installed with a seal (e.g. Sikaflex), up to IP66 can be achieved.



4.4 Wiring of the Panda iControl3 control unit

Fig. 4.4-1: Wiring of the Panda iControl3 control unit



The Panda iControl3 control unit is connected to the wire harness with a 42-pin socket, see “Terminal assignment of the 42-pin socket” on page 32.

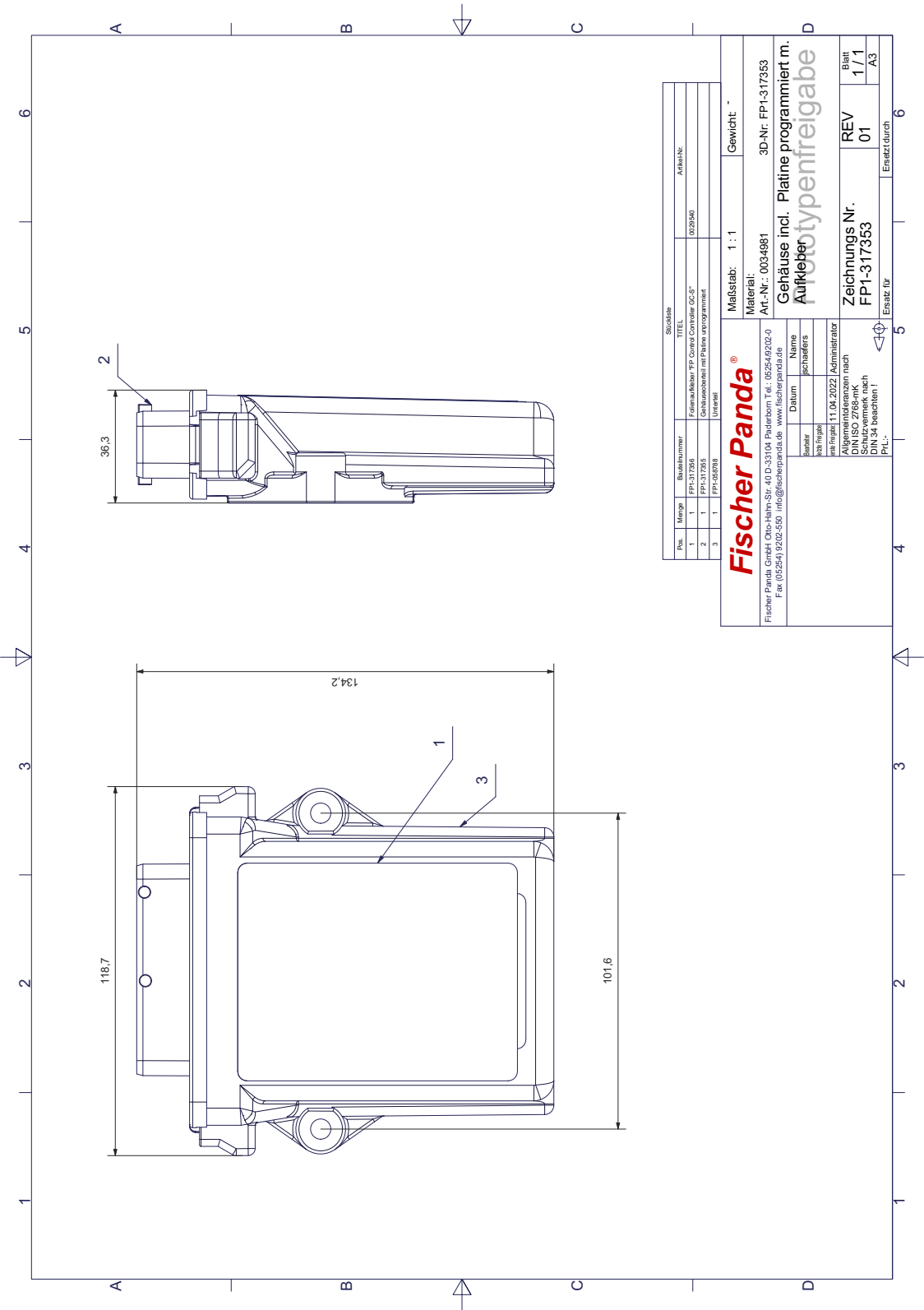
4.4.1 Terminal assignment of the 42-pin socket

Fig. 4.4.1-1: Terminal assignment of the 42-pin socket

Terminal	IN / OUT	Function
1	OUT	Glow plug(s)
2	IN	Spare input
3	--	not used
4	--	Supply voltage Inverter
5	--	not used
6	OUT	Water pump
7	OUT	Booster (optional, depending on generator type)
8	--	not used
9	--	GND
10	--	CAN high 2
11	--	VCCS +5V servo motor (red wire)
12	--	GND
13	--	GND
14	--	Supply voltage+
15	OUT	Electric starter
16	IN	Spare input
17	IN	Cylinder head temperature
18	IN	Exhaust manifold temperature
19	IN	Oil pressure switch
20	IN	Universal input
21	IN	Emergency stop+
22	--	CAN high 1
23	--	CAN low 2
24	--	Bus supply voltage
25	--	GND servo motor (black wire))
26	OUT	Potential-free output (PWM-)
27	IN	Pick-up sensor
28	--	Supply voltage+
29	OUT	Fuel pump
30	IN	Spare input
31	IN	Spare input
32	IN	Winding temperature
33	--	GND
34	--	GND
35	IN	Emergency stop-
36	--	CAN low 1
37	IN	Wake up
38	--	GND
39	--	PWM servo motor (yellow wire)
40	OUT	Potential-free output (PWM+)
41	--	not used
42	--	GND

4.5 Dimensions Panda iControl3 control unit

Fig. 4.5-1: Housing of the Panda iControl3 control unit



With the iControl3 panel it is possible to operate up to four panels on one iGenerator (one master panel and three slave panels).

On the master panel the terminating resistor for the CAN bus must be switched on. The terminating resistor of the slave panel must be switched off. To do this there is a double DIP switch on the backside of the iControl3 panel. **See “Function of the double DIP Switch” on page 29.**

In an iControl system with master and slave panels, the master panel must always be the last panel in the row so that the terminating resistor is at the end of the CAN bus. Slave panels must be connected between the iControl3 controller and the master panel. **See “Master/Slave panel configuration” on page 34.**

Note! iControl2 system

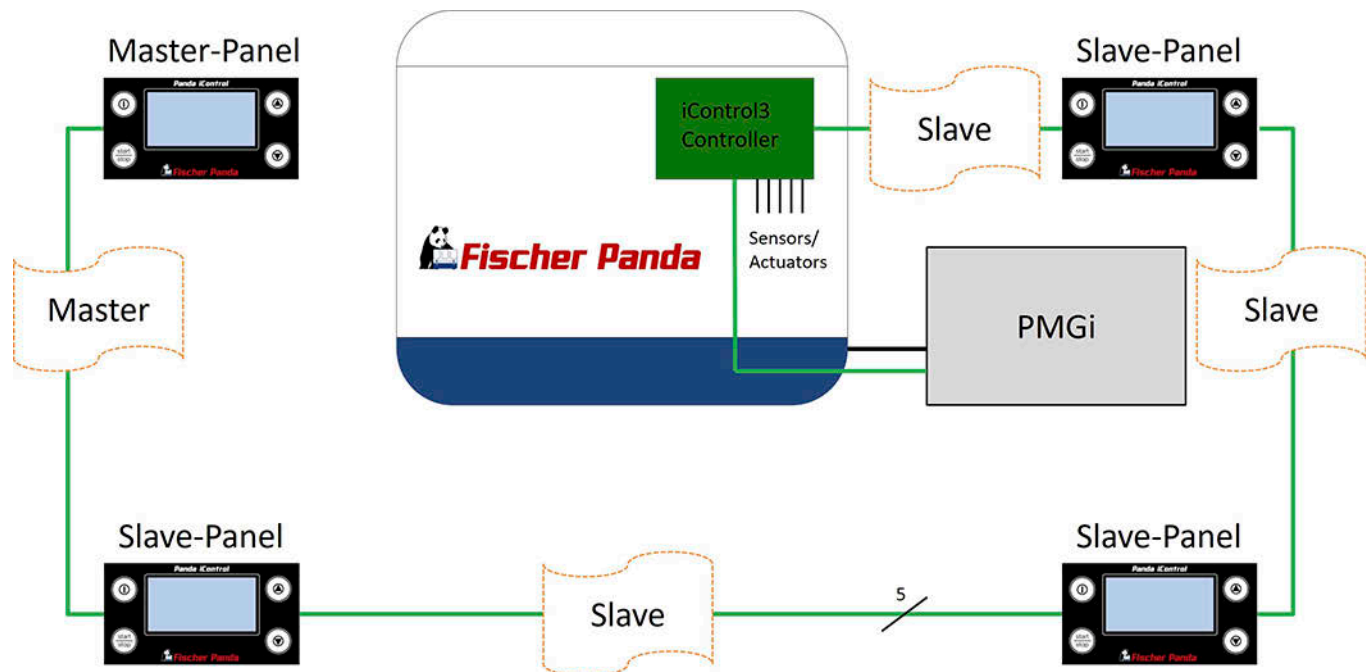


All iControl panels have the default network address "1". This address can be changed in the set up menu. Possible addresses are 1, 2, 3 and 4. **Each panel must have its own address. See "Setting the network ID" on page 18.**

Note! Autostart function



Fig. 4.6-1: Master/Slave panel configuration



4.7 Start-up

After completing the installation, the system must be started up.

For this purpose, the start-up log for the generator is processed and filled in by the specialist installing the equipment. The completed log shall be handed over to the operating company.

The operating company shall be instructed in the operation, maintenance, and hazards of the generator. This applies to both the maintenance steps and hazards described in the manual and to additional steps and hazards that result from the specific installation conditions and the connected components.

The original start-up log of the generator must be sent to Fischer Panda to obtain the full warranty. Make sure that you retain a copy for your records.

Note!



The corresponding forms are included in the generator manual.

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5. Maintenance

5.1 Maintenance of the iControl3 controller

The iControl3 controller is maintenance-free. The fuses of the controller are self-healing.

5.1.1 Cleaning the iControl3 controller

The housing shall be cleaned within the scope of the overall generator cleaning. The housing can be wiped off with a soft, lightly dampened cloth. In doing so, it must be ensured that no moisture enters the jacks and the housing.

5.2 Maintenance of the iControl3 remote control panel

The iControl3 remote control panel is maintenance-free.

5.2.1 Cleaning the iControl3 controller

The display can be cleaned with a soft cloth dampened lightly with soapy water. Harsh cleaning agents are not suitable and can cause the display film to turn dull.

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6. Warnings and error messages

To enable the safe operation of the generator, the Panda iControl3 controller is programmed with a series of warnings and error messages that influence the generator operation.

6.1 Warnings

Warnings are issued when the variable being monitored, e.g. temperature, reaches a defined warning threshold. The warnings are issued on the Panda iControl3 panel display via the cyclical display of the word „high“ or „low“, alternating with the measured value, e.g. the temperature. Warnings do not become active until the time between reaching the threshold value and the defined delay has expired.

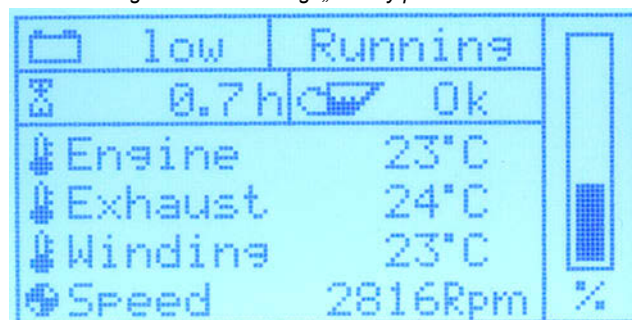
Warnings do not result in the generator or the controller being switched off. Note!



6.1.1 Examples of warnings on the display:

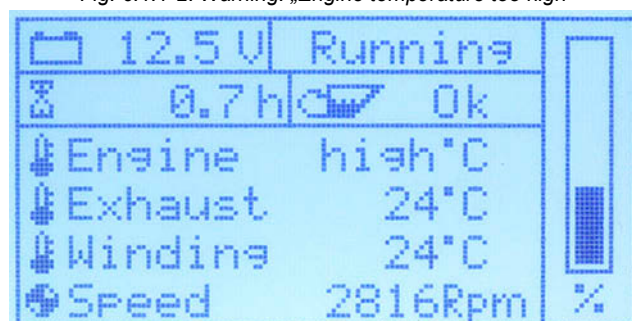
Warning: „Battery power too low“

Fig. 6.1.1-1: Warning: „Battery power too low“



Warning: „Engine temperature too high“

Fig. 6.1.1-2: Warning: „Engine temperature too high“



6.1.2 Warning messages

All warning messages defined for the Panda iControl3 and the corresponding display output are compiled in the subsequent table.

Fig. 6.1.2-1: Warning messages

Warning message on the display	Meaning of warning message
„high“ flashes, alternating with the temperature value of the cylinder head	Cylinder head temperature is too high, the warning threshold was reached
„high“ flashes, alternating with the temperature value of the winding	Winding temperature is too high, the warning threshold was reached
„high“ flashes, alternating with the temperature value of the exhaust manifold	Exhaust manifold temperature is too high, the warning threshold was reached
„low“ flashes, alternating with the voltage value of the starter battery	Starter battery voltage is too low, the warning threshold was reached

6.2 Faults

Error messages are issued when the monitored variable, e.g. a temperature, reaches the defined fault threshold.

With the temperature sensors, a loose connection or a broken cable will result in a fault, as well, and cause the generator to shut down.

An error message is typically preceded by a warning, as the warning threshold is reached before the fault threshold. Error messages are output on the Panda iControl3 panel display in the form of the error text shown on a cleared display page. Faults do not become active until the time between reaching the fault threshold and the defined delay has expired.

Faults result in the generator shutting down. If a fault occurs due to the battery voltage being too low, the controller is completely shut down to prevent the battery from discharging too much.

Examples of an error message on the display:

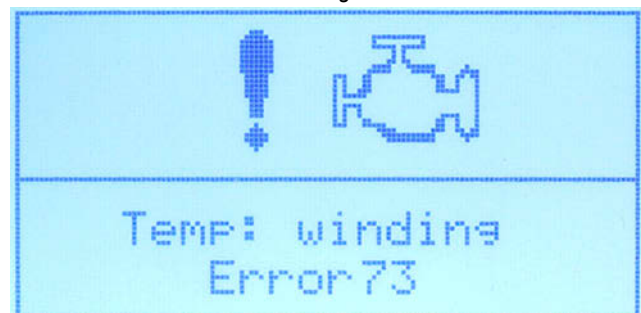
Fault: „Exhaust Error“, cable break at the temperature sensor of the exhaust manifold

Fig. 6.2-1: Fault: Cable break temp. sensor exhaust manifold



Fault: „Temp: Winding Error 73“, winding temperature too high

Fig. 6.2-2: Fault: „Temp: Winding Error 73“, winding temperature too high



6.2.1 Error messages

All error messages defined for the Panda iControl3 and the corresponding display texts are compiled in the subsequent table.

Fig. 6.2.1-1: Error messages

Error message on the display	Meaning of error message
„Error“ is output instead of a temperature	Broken cable on corresponding temperature sensor

Fig. 6.2.1-2: Error codes

Error code	Meaning	Error Message English	Error Message German
1	Inverter voltage low L1	L1 undervoltage	L1 Unterspannung
2	AC frequency low L1	L1 low frequency	L1 Frequenz niedrig
5	Emergency stop	Emergency stop	Nothalt
7	Oil pressure fault	FAULT: OILPRESS	Fehler: Öldruck
8	Engine temperature out of range	OUT: Cyl. head	OUT: Motor
9	Winding temperature out of range	OUT: Winding	OUT: Wicklung
10	Exhaust temperature out of range	OUT: Exhaust	OUT: Abgas
25	Starter battery voltage low	Battery voltage low	Batt. Unterspannung
26	Starting fails	Starting fails	Startfehler
28	unexpected stop (revolution speed low)	UNEXPEXED STOP	Unerwarteter stopp
30	Inverter voltage low L2	L2 undervoltage	L2 Unterspannung
31	AC frequency low L2	L2 low frequency	L2 Frequenz niedrig
34	Inverter voltage low L3	L3 undervoltage	L3 Unterspannung
35	AC frequency low L3	L3 low frequency	L3 Frequenz niedrig
65	Inverter voltage high L1	L1 Overvoltage	L1 Ueberspannung
66	AC frequency L1 high	L1 frequency high	L1 Frequenz hoch
67	Inverter overload L1	L1 Overload	L1 Ueberlast
70	Servo current high	Servo current high	Servo Ueberstrom
72	Engine temperature fault	Temp: cyl. head	Temp. Motor
73	Winding temperature fault	Temp: winding	Temp. Wicklung
74	Exhaust temperature fault	Temp: exhaust	Temp. Abgas
75	Electronic temperature high	ECU temp. high	ECU Temperatur hoch
77	Starter output high	Starter overcurrent	Anlasser Ueberstrom
78	Glow plugs output high	Glowplug overcurrent	Gluehkerze Ueberstrom
79	Flame plug output high	Flameplug overcurrent	Flammkerze Ueberstrom
80	Fuel supply output high	Fuel pump overload	Kraftstoffpumpe Ueberstrom
81	ETR-Hold current high	ETR-hold overload	ETR-hold Ueberstrom
82	ETR-Pull current high	ETR-pull overload	ETR-pull Ueberstrom
83	Fan output high	Fan overcurrent	Luefter Ueberstrom
85	Boost relay high	Boost overcurrent	Boost Ueberstrom
86	Bus current high	Bus overcurrent	Bus Ueberstrom
89	Starter battery voltage high	Battery voltage high	Batt. Ueberspannung
90	Engine revolution (RPM) high	Overspeed	Ueberdrehzahl
94	Inverter voltage high L2	L2 Overvoltage	L2 Ueberspannung
93	Power contactor	Inv. switch overload	Inv. Schalter Ueberlast
95	AC frequency L2 high	L2 frequency high	L2 Frequenz hoch
96	Inverter overload L2	L2 Overload	L2 Ueberlast
98	Inverter voltage high L3	L3 Overvoltage	L3 Ueberspannung
99	AC frequency L3 high	L3 frequency high	L3 Frequenz hoch

Error code	Meaning	Error Message English	Error Message German
100	Inverter overload L3	L3 Overload	L3 Ueberlast
103	Universal output 1 high	UOut1 overcurrent	UOut1 Ueberstrom
104	Universal output 2 high	UOut2 overcurrent	UOut2 Ueberstrom
130	CAN communication lost	BUS ERROR	Bus Fehler
135	Synchronisation problem parallel inverters	Inv. sync. fails	Inv. sync. Fehler
136	CAN communication to external ECU lost	ECU com. lost	ECU Komm. verloren
138	Diagnostic Message (ECU)	Engine fault	Motorstoerung
142	Red Stop Lamp (RSL)	Engine fault	Motor Fehler
148	Error three-phase rotating field	Error field rotation	falsches Drehfeld
150	Init failed (no generator type is selected)	Init fail	Init fail
151	Watchdog Error	Watchdog	Watchdog
152	Inverter temperature high L1	L1 temperature high	L1 Temperatur hoch
153	Inverter temperature high L2	L2 temperature high	L2 Temperatur hoch
154	Inverter temperature high L3	L3 temperature high	L3 Temperatur hoch
155	Inverter temperature high DC link	DC temperature high	DC Temperatur hoch
163	Inverter DC link overload	DC Overload	DC Ueberlast
164	Inverter DC link overvoltage	DC Overvoltage	DC Ueberspannung

Error messages can be acknowledged with the start/stop button, The controller will then return to stand-by mode.

6.2.2 Warning and fault thresholds

The threshold values resulting in triggering warnings and faults depend on the generator type and are compiled in table below.

Fig. 6.2.2-1: Warning and fault thresholds for different generator types

Generator type	Warning/fault	Warning threshold	Fault threshold
5000i marine	Cylinder head temperature	85 °C	95 °C
	Delay	5 s	5 s
	Winding temperature	130 °C	135 °C
5000i vehicle	Delay	5 s	5 s
	Exhaust manifold temperature	70 °C	75 °C
	Delay	1 s	1 s
P8000i / P10000i marine	Cylinder head temperature	90 °C	95 °C
	Delay	5 s	5 s
	Winding temperature	130 °C	135 °C
P8000i / P10000i vehicle	Delay	5 s	5 s
	Exhaust manifold temperature	70 °C	75 °C
	Delay	1 s	1 s

Generator type	Warning/fault	Warning threshold	Fault threshold
P8-P50 marine	Cylinder head temperature Delay	90 °C 5 s	95 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	70 °C 1 s	75 °C 1 s
P8-P50 vehicle	Cylinder head temperature Delay	95 °C 5 s	100 °C 5 s
	Winding temperature Delay	160 °C 5 s	165 °C 5 s
	Exhaust manifold temperature Delay	100 °C 1 s	105 °C 1 s
P15000i marine	Cylinder head temperature Delay	90 °C 5 s	95 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	70 °C 2 s	75 °C 2 s
P15000i vehicle	Cylinder head temperature Delay	90 °C 5 s	95 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	95 °C 2 s	100 °C 2 s
P25i Marine	Cylinder head temperature Delay	90°C 5s	95°C 5s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	70°C 2 s	75°C 2 s
P25i vehicle	Cylinder head temperature Delay	90 °C 5 s	95 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	100 °C 2 s	105 °C 2 s
P45i marine 230V/400V	Cylinder head temperature Delay	90 °C 5 s	95 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	80 °C 2 s	85 °C 2 s
P45i vehicle 230V/400V	Cylinder head temperature Delay	98 °C 5 s	105 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	100 °C 2 s	105 °C 2 s
P45i marine 3x230V	Cylinder head temperature Delay	98 °C 5 s	105 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	70 °C 2 s	75 °C 2 s

Generator type	Warning/fault	Warning threshold	Fault threshold
P45i vehicle 3x230V	Cylinder head temperature Delay	98 °C 5 s	105 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	100 °C 2 s	105 °C 2 s
P60i marine	Cylinder head temperature Delay	90 °C 5 s	95 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	70 °C 2 s	75 °C 2 s
P60i vehicle	Cylinder head temperature Delay	90 °C 5 s	95 °C 5 s
	Winding temperature Delay	130 °C 5 s	135 °C 5 s
	Exhaust manifold temperature Delay	95 °C 2 s	98 °C 2s
All generator types	Starter battery voltage low Delay	11.8 V 30 s	10.8 V 30 s
	Starter battery voltage high	15.0 V 5 s	-- --

6.2.3 Bus errors

If the communication connection is lost on the Fischer Panda bus, an error is output on the display after a period of 10 seconds:

This error will occur if at least one of the two data lines of the Fischer Panda bus is disconnected. Once the connection is restored, the error message can be acknowledged with the start/stop button.

Fig. 6.2.3-1: Error in the communication (Fischer Panda bus)



If the communication connection is lost, the generator shall be secured (open battery disconnect switches), and all plug-in connections and cables shall be checked for firm seating or damage.

7. Annex

7.1 Technical data

7.1.1 Technical data for iControl3 control unit

Fig. 7.1.1-1: Technical data for iControl 3 control unit

	iControl3 control unit
Supply voltage	12 V-13,5 V (12 V automotive)
Current consumption, nominal	80 mA @14,4 V
Current consumption, stand-by	< 0,1 mA
Operating temperature	-20 °C to +105 °C
Storage temperature	-30 °C to +125 °C
Housing	Automotive, PBT GF30
Protection class	IP65
Overall dimensions	134,2 x 118,7 x 36,3 mm (L x W x H including connector)
Connector	42-pin connector
Weight	0,25 kg
FP part number	0034981

7.1.2 Technical data for iControl3 remote control panel

Fig. 7.1.2-1: Technical data for iControl3 remote control panel

	iControl3 control unit
Supply voltage	12 V-24 V (12 V or 24 V automotive)
Current consumption, off	0 mA
Current consumption, stand-by - backlight brightness 90 %	60 mA
Current consumption, stand-by - backlight brightness 40 %	50 mA
Current consumption, stand-by - backlight brightness 0 %	40 mA
Operating temperature	-20 °C to +70 °C
Storage temperature	-30 °C to +80 °C
Housing	ABS plastic
Protection class	IP04
Overall dimensions	120 x 65 x35 mm (L x W x H), Cutout: 109,2 x 54,5 mm
FP part number	0034806

7.2 CO₂ balance derived from the emission measuring cycle for engines in accordance with 2016/1628 EC

The following CO₂ balance derived from the emission measuring cycle is applicable, with regard to the engine, to generators that are approved in accordance with 2016/1628 EC:

Fig. 7.2-1: CO₂ balance derived from the emission measuring cycle for engines in accordance with 2016/1628 EC

CO ₂ balance derived from the emission measuring cycle				
Engine	Engine Category	Engine family type	Type approval	CO ₂ balance - Test cycle [g/kwh]
Z482	NRE-v-2	HKBXL.778KCB	e1*2016/1628*2016/1628EV2/D*0008*00	1019.8
D722	NRE-v-2	HKBXL.778KCB	e1*2016/1628*2016/1628EV2/D*0008*00	
Z602	NRE-v-2	HKBXL.898KCB	e1*2016/1628*2016/1628EV2/D*0009*00	1047.4
D902	NRE-v-2	HKBXL.898KCB	e1*2016/1628*2016/1628EV2/D*0009*00	
D1105	NRE-v-2	HKBXL01.5BCB	e1*2016/1628*2016/1628EV2/D*0010*04	1018.0

The emission decal on the valve cover indicates the emission homologation to which the engine belongs.

Fig. 7.2-2: Example Z482 E4B IMS2

